

## ***B. Tech Degree VI Semester Special Supplementary Examination November 2012***

### **ME 601 INSTRUMENTATION AND CONTROL SYSTEMS (2006 Scheme)**

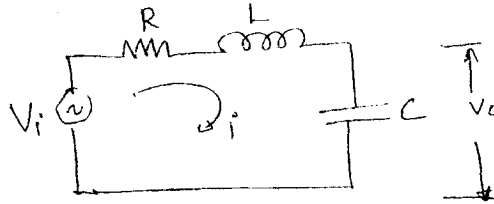
Time : 3 Hours

Maximum Marks : 100

#### **PART A** (Answer **ALL** questions)

(8 x 5 = 40)

- I. (a) What are the different functional elements of an instrument?  
(b) Write a short note on static and dynamic characteristics of errors.  
(c) State and explain five laws of thermocouple.  
(d) What do you mean by calibration? Also explain the term sensitivity.  
(e) Define transfer function. Give the transfer function of the given networks.



- (f) Give force voltage analogy and force current analogy.  
(g) Explain BIBO stability. State Routh-Hurwitz criterion.  
(h) State and explain Nyquist criterion.

#### **PART B**

(4 x 15 = 60)

- II. (a) Explain the different classification of errors in measurements. (12)  
(b) What is a transducer? (3)

**OR**

- III. Write a short note on generalized mathematical modelling of measuring systems. What do you mean by first order, second order and third order instruments? (15)  
IV. Explain the working principle of a strain gauge. Explain the construction of bonded and unbonded strain gauge. Derive an expression for gauge factor. (15)

**OR**

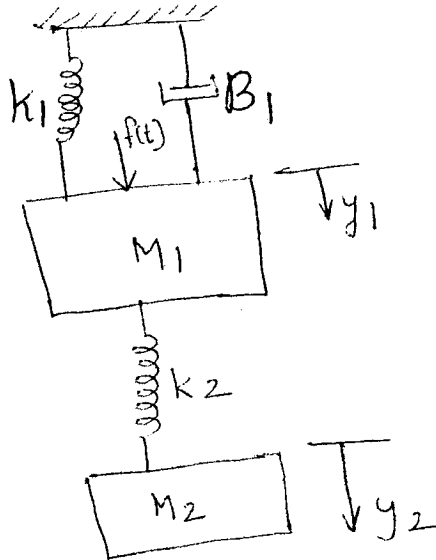
- V. (a) Explain the working and constructional details of an optical pyrometer. (10)  
(b) Give the working principle of an RTD. (5)  
VI. Explain the transient response of a second order control system. Also derive expressions for different time domain specifications. (15)

**OR**

(P.T.O)

VII. (a) What are the different static error constants? Explain. (5)

(b) Determine  $\frac{Y_2(s)}{F(s)}$  for the system shown. (10)



VIII. Sketch Bode – plot for the given transfer function. Also determine gain crossover frequency and phase margin. (15)

$$G(s)H(s) = \frac{10(S+50)}{S(S+5)}$$

OR

IX. A certain unity negative feed back system has the following open loop transfer function (15)

$$G(s)H(s) = \frac{K}{S(S+1)(S+3)}. \text{ Draw root locus for } 0 \leq K \leq \alpha.$$